THE FUNDAMENTAL THEOREM OF ALGEBRA: THE COMPLEX NUMBERS ARE ALGEBRAICALLY CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The proof uses the intermediate value theorem, i.e. an analytic input, as the author notes in the section opening.

Editorial note. The label lemma-C-algebraically-closed names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Revision r002: regenerated from the same spec after the r001 assembly failed at compile time (the r001 context span ran through line 3422 and overlapped the statement span 3418-3421, so the lemma environment and its label were injected twice and the engine reported multiply-defined labels; the context now ends at line 3416, before the statement environment begins; the stale r001 job directory is retained as evidence).

Allowed standard prerequisites (editorial).

- The intermediate value theorem from calculus: every odd degree real polynomial has a real root, by sign change of the dominating leading term
- Galois theory for finite extensions: for a finite Galois extension with group G and a subgroup P, the fixed field K^P has degree over the base equal to the index of P in G
- The 2-sylow theorem: every finite group has a 2-sylow subgroup, and the index of a 2-sylow subgroup is odd
- Up to isomorphism the only degree two extension of the reals is the complex numbers, and every complex number has a square root, so the complex numbers admit no quadratic extension
- Fields, Lemma lemma-normal-closure: every finite extension can be embedded in a finite normal extension over the base

Source reference. Original reference 09DT: sources/stacks-fields/pinned-original-source.tex, source lines 2303–2303. Prerequisite pointer invoked at 3469; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

1. The complex numbers

The fundamental theorem of algebra states that the field of complex numbers is an algebraically closed field. In this section we discuss this briefly.

The first remark we'd like to make is that you need to use a little bit of input from calculus in order to prove this. We will use the intuitively clear fact that every odd degree polynomial over the reals has a real root. Namely, let $P(x) = a_{2k+1}x^{2k+1} + \ldots + a_0 \in \mathbf{R}[x]$ for some $k \geq 0$ and $a_{2k+1} \neq 0$. We may and do assume $a_{2k+1} > 0$. Then for $x \in \mathbf{R}$ very large (positive) we see that $P(x) > 0$ as the term $a_{2k+1}x^{2k+1}$ dominates all the other terms. Similarly, if $x \ll 0$, then $P(x) < 0$ by the same reason (and this is where we use that the degree is odd). Hence by the intermediate value theorem there is an $x \in \mathbf{R}$ with $P(x) = 0$.

A conclusion we can draw from the above is that $\mathbf{R}$ has no nontrivial odd degree field extensions, as elements of such extensions would have odd degree minimal polynomials.

Next, let $K/\mathbf{R}$ be a finite Galois extension with Galois group G. Let $P \subset G$ be a 2-sylow subgroup. Then $K^P/\mathbf{R}$ is an odd degree extension, hence by the above $K^P = \mathbf{R}$, which in turn implies $G = P$. (All of these arguments rely on Galois theory of course.) Thus G is a 2-group. If G is nontrivial, then we see that $\mathbf{C} \subset K$ as $\mathbf{C}$ is (up to isomorphism) the only degree 2 extension of $\mathbf{R}$. If G has more than 2 elements we would obtain a quadratic extension of $\mathbf{C}$. This is absurd as every complex number has a square root.

The conclusion: $\mathbf{C}$ is algebraically closed. Namely, if not then we'd get a nontrivial finite extension $K/\mathbf{C}$ which we could assume normal (hence Galois) over $\mathbf{R}$ by Lemma 09DT. But we've seen above that then $K = \mathbf{C}$.

Lemma 1.1 (Fundamental theorem of algebra)**.** *The field $\mathbf{C}$ is algebraically closed.*

Proof. See discussion above. □